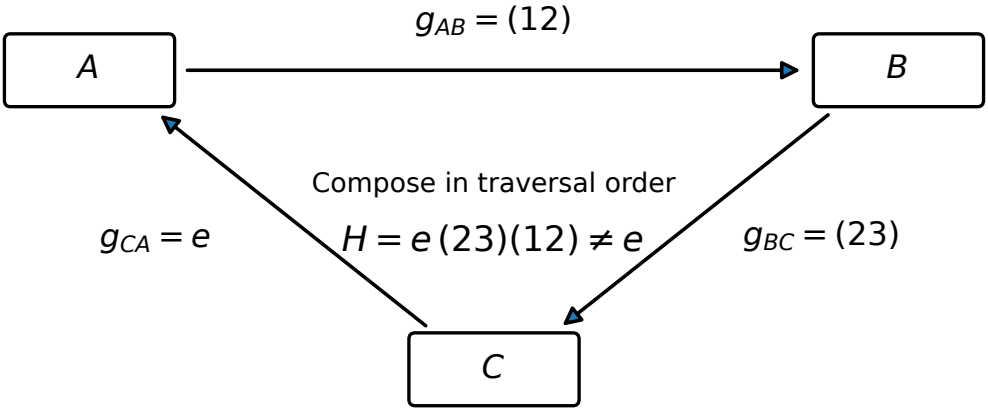

Every edge is invertible; the round trip is not identity.